

DESKTOP-DOM: ARCHITECTURAL THINKING & SYSTEM SPAN

Kernel Accessibility, Sub-Millisecond Entity Disambiguation, & The Crclre Intent-to-Action Execution Zone

The Fundamental Thesis: Vision-based computer-use agents (Claude, Operator) fail on consumer desktops: treating the screen as 8M raw pixels burns ~2,000 tokens/step, introduces 3–5s latency, drifts on Retina coordinates, and physically hijacks the cursor. **Desktop-DOM** queries native OS accessibility buses (AXUIElement / UIAutomation) in **<15ms** with an **88% token reduction** and zero cursor displacement. Paired with **Aura** (the minimalist native intent pill), it delivers the backend foundation for Crclre's core mission: *The Intent Layer of Computing*.

1. Evolutionary Multi-Tier Architecture (Layer 0 to Layer 3 + Level 2.5 Execution Zone)

Layer 0 Kernel Bus	Native OS Accessibility PyObjC, Win32, AT-SPI2	Traverses accessibility trees via AXUIElementCopyAttributeNames / kAXChildrenAttribute. Extracts exact bounding boxes, roles, and states without capturing pixels. Background action dispatch via kAXPressAction and Quartz HID.	8–18 ms 0px cursor move
Layer 1 Fast-Path	Deterministic Router & Scanner AST Pruner, SequenceMatcher	Prunes 88% of redundant non-semantic structural nodes. Indexes 100+ native apps dynamically across /Applications. Resolves typos (spotify→Sotify) via SequenceMatcher (≥0.68). Native folder navigation & safe app quit.	<25 ms Zero LLM tokens
Layer 2 Intent Core	Personal Memory & Local Model SQLite WAL + Mistral SLM	Where Crclre.ai fundamentally lies: Resolves colloquial human intent (<i>message Josh</i>) via 5-tier disambiguation (<0.5ms). Grounded local Mistral model ingests active desktop DOM + SQLite personal entity graph to translate human intent to deterministic actions.	0.49 ms (Mem) 720 ms (Mistral)
Layer 2.5 Exec Zone	Ghost Cursor & State Diffing 3-Tier Cursor-Free Engine	The Execution Zone between L2 & L3: Gives AI its own virtual cursor without disturbing user's mouse. Tier 1: kAXPressAction in-memory. Tier 2: CGWarpMouseCursorPosition warp-and-restore in <0.8ms via KCGHIDEventTap. Tier 3: kAXValueAttribute mutation. \$O(N)\$ DOM diffing (\$T_1 - T_0\$) verifies state in <0.25ms.	<0.8 ms (Ghost) <0.25 ms (Diff)
Layer 3 Agentic Loop	Autonomous ReAct Loop Multi-turn Planning & Daemon	Full multi-step autonomous planning with self-reflection. Headless HTTP daemon (desktop-dom serve on port 8484) exposing tree, intent, diff, and agent endpoints for seamless integration into Crclre's proprietary frontend.	Multi-turn Headless daemon

2. Deep Technical Systems: Syscalls, Memory Layouts, IPC & Empirical Benchmarks

Kernel Syscall & IPC Bridges:

- **WebKit → Cocoa → Python:** Omnibar uses WKScriptMessageHandler posting JSON to Python event loop; dispatches in <1.2ms without HTTP overhead.
- **Ghost Cursor Syscall Path:** CGEventGetLocation → CGEventCreateMouseEvent → CGEventPost(KCGHIDEventTap) → instant CGWarpMouseCursorPosition(cur_pos) in <0.8ms.
- **Direct In-Memory Mutation:** AXUIElementSetAttributeValue(kAXValueAttribute, val) populates text fields without synthetic keystroke jitter or stealing window focus.

Memory Hierarchy & Token Compression:

- **Personal Entity Graph:** SQLite in WAL mode with dual-layer memory cache (~./aura/memory.db). 5-tier fuzzy ranking resolves in **0.49ms**.
- **Zero-Disk Audio Pipeline:** 16kHz float32 circular NumPy buffer feeds local Whisper directly from RAM with zero disk I/O.
- **AST Token Pruning:** \$O(N)\$ cycle-safe DFS eliminates non-semantic nodes, shrinking 2,000 raw nodes to ~220 actionable tokens (**88% compression**).

Dimension	Vision-Only (Claude/Gemini)	Desktop-DOM (Aura)
Tree / State Capture	80–150ms (4K PNG encode)	8–18ms (Native OS bus)
Token Overhead	1,500–2,200 tokens (\$0.03)	180–350 tokens (88% saved)
Execution Latency	3,000–5,000ms per step	<25ms (Fast) / 720ms (SLM)
Cursor Stability	Steals physical mouse cursor	Cursor-Free Ghost Cursor
Retina / Scaling	Prone to DPI fractional drift	Exact OS Bounding Boxes
Offline Privacy	Requires cloud GPU inference	100% Local & Private

3. Production Verification Ledger & Founder Alignment

115 / 115 Passing (100%) Zero flaky tests in ~21s. Full coverage: schema, pruners, memory, diff, agent, fast-paths.	SQLite in WAL Mode Dual-layer in-memory hot cache. 5-tier disambiguation executes in 0.49ms with zero cloud lock.	Sub-ms Headless Daemon desktop-dom serve on port 8484 exposes tree, intent, diff, and agent endpoints for Crclre.	Joshua & Cyril Rayan Delivers the exact high-velocity backend engine solving Crclre's core intent layer bottlenecks.